

Issues for temporal counterpart theory

Handout for *Time and Modality* seminar

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1 Ontology for supersubstantialist temporal counterpart theorists

Supersubstantialism: every ordinary object is identical to some region of spacetime. Standard B-theoretic ‘worm’ supersubstantialism.

Three questions for temporal counterpart theorist supersubstantialists: (i) Which regions are the ordinary objects? (ii) What does it take for regions to fall under ordinary 3D geometrical predicates? (iii) Which regions are past- and future-counterparts of which; and more generally, which permutations are counterpartings?

- (A) The stage view. Ordinary objects are achronal. 3D geometric predicates work in the obvious way—in particular, x is spatially within y iff x is a subregion of y . Temporal counterparthood is highly context sensitive: if we are thinking of an object as a statue we will resolve the context sensitivity one way, while if we are thinking of the same object as a lump we will resolve it a different way. Distinct ordinary objects never spatially coincide.
- (B) The growing worm view. Ordinary objects are initial segments of the B-theorists’ worms. 3D geometric predicates depend on the spacetime geometric features of the objects’ leading edges. The past counterparts of an ordinary object are its initial segments. Distinct ordinary objects can coincide, but only when they have failed to coincide in the past.
- (C) The shrinking worm view. Ordinary objects are final segments of the B-theorists’ worms. 3D geometric predicates depend on the spacetime geometric features of the objects’ trailing edges. The future counterparts of an ordinary object are its final segments. Distinct ordinary objects can coincide, but only when they will fail to coincide in the future.
- (D) The ordered pair view (not actually a form of supersubstantialism, but. . .). Ordinary objects are ordered pairs whose first member is a complete worm and whose second member is a hypersurface of simultaneity. 3D geometric predicates depend on the spacetime geometric features of the intersection of the two elements. The temporal counterparts of an ordinary object are ordered pairs with the same first member and different (earlier or later) second members. Distinct ordinary objects can coincide, but they cannot permanently coincide.
- (E) First way of coding ordered pairs as single regions: code $\langle w, s \rangle$ as the *sum* (least upper bound) of w and s . Recover s as the largest hypersurface part, and w as the interior (or the closure of the interior. . .).
- (F) Second way of coding ordered pairs as single regions: code $\langle w, s \rangle$ as w/s . Recover w as the closure (or the interior of the closure), and s as the unique hypersurface that does not overlap but overlaps the interior of the closure. . .

2 Ontology for Butlerian temporal counterpart theorists

Bishop Butler: Each thing is what it is, and not another thing.

One version: ordinary objects have fundamental objects as parts. There is a certain range of “categorical”, non-modal, non-temporal predicates that include 3D geometric predicates, such that distinct objects with the same fundamental parts agree on these predicates.

In this setting, the pointy manifold picture requires us to say that ordinary objects have spacetime points as parts. Which ones? The most natural thought: the points that compose some *stage*.

Question: are there ordinary objects such that *necessarily* they have exactly the same fundamental parts? Arguably yes: Fine’s letters. But let’s say no for now.

How do temporal counterpartings act on a given ordinary object? This is going to be determined by two things: (i) how do temporal counterpartings act on fundamental objects (points?)? (ii) What are the necessary and sufficient conditions for a fundamental object to be part of the ordinary object in question?

How do temporal counterpartings act on spacetime points? Newtonian spacetime: easy. Neo-Newtonian spacetime: worry about arbitrariness. Resolve this worry by appeal to vagueness.

- If we don’t want the systematic vagueness in ‘What’s going to happen to the points?’ to percolate up into systematic vagueness in ‘What’s going to happen to ordinary objects such as planets?’, we need to make sure that the modal profiles of ordinary objects are invariant under certain spatial shifts (explain).

3 Sider on fission cases

Past Totality $(P\phi \wedge P\psi) \rightarrow (P(\phi \wedge P\psi) \vee P(\phi \wedge \psi) \vee P(\psi \wedge P\phi))$

Future Totality $(F\phi \wedge F\psi) \rightarrow (F(\phi \wedge F\psi) \vee F(\phi \wedge \psi) \vee F(\psi \wedge F\phi))$

In the standard model theory these correspond to the following conditions:

(PT) If $t_1 < t_3$ and $t_2 < t_3$, then either $t_1 < t_2$ or $t_1 = t_2$ or $t_2 < t_1$.

(FT) If $t_3 < t_1$ and $t_3 < t_2$, then either $t_1 < t_2$ or $t_1 = t_2$ or $t_2 < t_1$.

Under COUNTERPAIRINGS they correspond to the following constraints on counterpartings:

(PT) If R and S are past-counterpairings, then either $R \circ S^{-1}$ or $S \circ R^{-1}$ is a past-counterpairing, or $R = S$.

(FT) If R and S are future-counterpairings, then either $R \circ S^{-1}$ or $S \circ R^{-1}$ is a future-counterpairing, or $R = S$.

The scenario: A person—call him Fred—has gone into a doctor’s office and will soon be subjected to a brain bisection and double transplantation. . . Afterwards, one of the two donor bodies will live in the lap of luxury, while the other will be out on the street. What’s

in store for Fred?

- (i) Parfit: Fred will never be rich, and will never be poor. In fact he will soon cease to exist. There will be a rich person and a poor person who are like Fred in certain respects, but neither of these people will *be* Fred.
- (ii) Fred will be rich *and* poor. He will be in two places at once, thinking two sets of thoughts, and unable to introspectively access his total package of experiences.
- (iii) Williamson: Either Fred will be rich and will never be poor, or else he will be poor and will never be rich. In addition to Fred, there will one other person around who seems to remember Fred's childhood, etc., but in fact came into existence at the time of the operation.
- (iv) Lewis: Strictly speaking, there are (at least) *two* people now standing in the doctor's office. So far, they have always coincided. But they will soon cease to coincide. One of them will later be rich, and one of them will later be poor. No new people will be created when the operation is performed.
- (v) Sider: Fred will be rich. Also, Fred will be poor. But Fred will never be both rich and poor. Nor will he ever be in two places at once, or unusually lacking in introspective access to his total package of experiences. He will always be a psychologically ordinary, spatially unified person.

The scenario: Ed and Ted have just emerged from one of those operations. What were Ed and Ted like 24 hours ago?

- (i) Neither Ed nor Ted was around at all: they recently were created with misleading memories. . . .
- (ii) Ed is Ted, and both were around 24 hours ago, and have been alive for many years.
- (iii) One of the two was recently created and wasn't around 24 hours ago, while the other has been alive for years.
- (iv) They have both been alive for many years; until recently they were coincident.
- (v) a. They have both been alive for many years; 24 hours ago they were identical, but now they are not any more.
b. Ed was waiting in the doctor's office 24 hours ago. Also, Ted was waiting in the doctor's office 24 hours ago. But it was not the case 24 hours ago that (Ed and Ted are both waiting in the doctor's office). Indeed, this has never been the case: it has always been the case that (if Ed is a young person, Ted isn't around at all, and if Ted is a young person, Ed isn't around at all). Thus, it's ALSO true that 24 hours ago, Ed wasn't around at all, and that 24 hours ago, Ted wasn't around at all.

4 Time travel